

# PAVAN SOMA SHEKAR

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## EDUCATION

Northeastern University, Boston, MA  
Master of Science in Information Systems

May 2024

SJB Institute of Technology, Bengaluru, India  
Bachelor of Engineering in Computer Science and Engineering

Jul 2019

## WORK EXPERIENCE

Associate Software Engineer, Red Hat, Raleigh, NC

Nov 2024 – Present

- Delivered **upstream contributions** across **Foreman/Katello**, owning **full-stack features**, **bug fixes**, and **maintainability improvements** across **React**, **Ruby on Rails**, **databases**, and **automated tests**, with **downstream** impact on **Red Hat Satellite**
- Enhanced Flatpak, container image, repository, subscription, content view, and host workflows by delivering **UI**, **backend**, **database**, and **test changes** that improved **usability**, **reliability**, and **regression coverage**
- Improved **performance** and **upgrade reliability** through database-backed workflow improvements, **bulk SQL processing**, **parallelized migration handling**, and **multi-threaded execution fixes**, improving large-scale data accuracy
- Modernized **quality engineering** by contributing to **Robottelo/Airgun** using **pytest**, stabilizing frontend regression coverage, and developing a **Playwright E2E testing POC** to enable **shift-left testing**, establish reusable test infrastructure, and improve release confidence
- Strengthened upstream quality by authoring **documentation** and **release notes**, **reviewing code**, supporting **technical discussions**, **delivering demos** to stakeholders, maintainers, QE, and open-source contributors and serving as **release owner** for **Katello 4.17.0, 4.17.1, 4.18.0, and 4.18.1**
- Advanced **AI enablement** by building a **RAG-based chatbot POC** for Red Hat Satellite, contributing to **Foreman MCP server** workflows, and creating an **AI quickstart tooling guide** to help developers adopt AI tools more effectively

Software Engineer, OnSolve, Bengaluru, India

Apr 2022 – Aug 2022

- Created the UI for the OnSolve platform, focusing on **responsiveness** and **user experience** using **React.js**
- Boosted the platform's performance by 15% by integrating 3 front-end modules and utilizing **React.js** for **lazy loading** and **code splitting** and **Redux** for **state management** and **memoization**, reducing load times
- Integrated **Spring Boot** backend with **React.js** front-end, enhancing API communication and boosting efficiency by 20%
- Implemented **unit tests** with **Jest** and **Enzyme** for frontend, and **JUnit** for backend, achieving 90%+ **code coverage**
- Ensured high quality through stringent **code reviews** and supported 2 OnSolve.UI releases via **CI/CD** with **GitHub Actions**, seamlessly integrating **Docker** for efficient delivery in containerized environments

Senior Systems Engineer, Infosys, Bengaluru, India

Sep 2019 – Apr 2022

- Drove innovation by developing **2 Angular**-based web applications with dynamic data binding and optimized API integration, improving backend performance with **Node.js/Express.js**, and reducing load times by 25%
- Formulated a comprehensive testing strategy utilizing **Protractor**, **Jasmine**, and **Karma** frameworks, achieving 90% test coverage for **end-to-end** and **unit testing**
- Spearheaded the InfyMansions project, developing **full-stack** modules with **MEAN** stack technologies and integrating **Material UI** for enhanced user interface design
- Engineered a high-accuracy **YOLOv3 chair detection model** for **Smart Classroom** using **OpenCV** and a **MEAN** stack-based dashboard, significantly enhancing the interactive learning environment
- Played a key role in workforce development by conducting workshops in **JavaScript**, **Angular**, and **React.js** for 200+ employees and **mentored** 10 new hires

## TECHNICAL SKILLS

- |                                          |                                                                                       |
|------------------------------------------|---------------------------------------------------------------------------------------|
| ● <b>Programming Languages</b>           | Java, Python, HTML, CSS, JavaScript, TypeScript                                       |
| ● <b>Web Frameworks &amp; Libraries</b>  | React.js, Angular, Node.js, Express.js, Spring Boot, Ruby on Rails, Django, Bootstrap |
| ● <b>Testing</b>                         | Jest, Enzyme, Protractor, Jasmine, Karma, Mocha, Chai, JUnit, pytest, Playwright      |
| ● <b>Databases &amp; Data Management</b> | MySQL, MongoDB, PostgreSQL, Redis, Kafka, Elasticsearch, Hibernate, Sequelize.js      |
| ● <b>Cloud</b>                           | AWS, Azure, GCP, Docker, CI/CD, Git, GitHub Actions, Packer, Terraform, Pulumi        |
| ● <b>AI &amp; Developer Tooling</b>      | RAG applications, MCP server workflows, AI developer tooling                          |

## CERTIFICATIONS

- AWS Certified Developer - Associate**, Validation Number - B1LNQCDJYM1EQ6WP
- Microsoft Certified AZ-900: Azure Fundamentals**, Credential ID - H576-2553